

Time-to-Digital Converter for RF Frequency Synthesis in 90 nm CMOS

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Abstract—We propose and demonstrate a 20 ps time-to-digital converter (TDC) in 90 nm digital CMOS. It is used as a phase/frequency detector and charge pump replacement in an all-digital PLL for a fully-compliant GSM transceiver. The TDC core is based on a pseudo-differential digital architecture that makes it insensitive to NMOS and PMOS mismatches. The time conversion resolution is equal to an inverter propagation delay, which is the finest logic-level regenerative timing in CMOS. The TDC is self calibrating with the estimation accuracy better than 1%. Measured INL is 0.7 LSB. The TDC consumes 1.3 mA from a 1.3 V supply.

Index Terms—All digital, calibration, compensation, deep-submicron CMOS, time-to-digital converter (TDC), frequency synthesizer.

I. INTRODUCTION

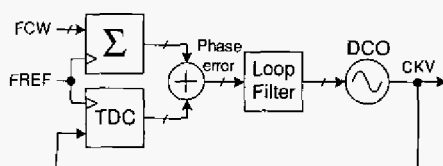


Fig. 1. TDC as a phase/frequency detector and charge pump replacement in an all-digital PLL (ADPLL)-based RF frequency synthesizer.

Time-to-digital converters (TDC's) are being widely used for time interval measurements in space science [1], high-energy physics [2] [3], laser range finders and test instrumentation [4]. Recently, TDC usage has been applied to frequency synthesis in DLL's for faster acquisition and to avoid false locking [5]. With the advent of digitally-intensive [6] and *all-digital PLL's* (ADPLL's) [7] in deep-submicron CMOS, the TDC is becoming an attractive replacement of the conventional phase/frequency detector and charge pump. This also allows to replace the loop filter that requires large and leaky integrating capacitors with a simple digital filter [8]. A very recent reference [9] culminates in a TDC-based frequency synthesizer for a fully-compliant GSM/EDGE transceiver in 90 nm CMOS. This paper covers the TDC architecture used in that design. The TDC is specifically employed there to measure edge time difference between the 1.6–2.0 GHz high-speed PLL *variable clock* (CKV) and the 26 MHz *frequency reference* (FREF) clock (Fig. 1), but it could also be used for a broader range of applications.

Older TDC architectures utilize an analog approach to first convert the time difference to voltage and then to

convert it to digital word by an ADC. More recent TDC architectures use multiphase ring oscillators and DLL's [5] to realize interpolation of clock edges, but the time resolution there is limited to a buffer delay. Other TDC architectures are based on a Vernier line [2] and pulse-shrinking techniques [1]. These two methods address the limitation of coarse intrinsic delay of buffer elements and apply the gate delay difference to improve the resolution below the sub-gate delay. However, the above approaches are analog intensive, require extensive calibration and suffer from long dead time, large power dissipation and large area.

The presented TDC architecture takes advantage of the deep-submicron CMOS process strengths of ultra-fast logic switching while avoiding its weaker handling of voltage resolution. It is recognized that an inverter propagation delay is the finest logic-level regenerative timing method in CMOS. To this extent, we avoid any control of the inverter delay that would make it slower, such as using current starving or increasing capacitive loads. For performance reasons, we also avoid fully differential structures that would require current biasing. The TDC resolution achieved here is equal to the fastest possible intrinsic inverter delay of 15-20 ps in this CMOS process, which is adequate for GSM-quality frequency synthesis [8][9]. The inverter-delay calibration method is fully digital and only involves processing of the TDC output.

The presented TDC system is also used as a built-in estimator of the process strength to set bias current of other analog circuits on the same SoC die. Indeed, the self-oscillating frequency of the 3-inverter ring oscillator structure is commonly used as a figure-of-merit (FOM) of the deep-submicron CMOS process strength.

II. BASIC TDC CORE ARCHITECTURE

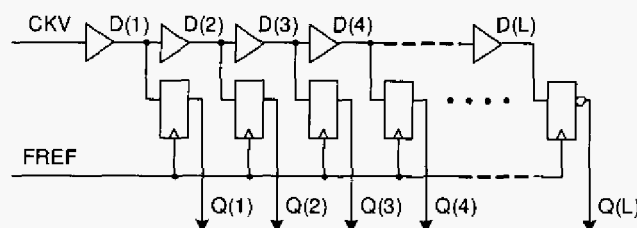


Fig. 2. Simple architecture of a time-to-digital converter (TDC) core.

Perhaps the simplest possible realization of a TDC core architecture is shown in Fig. 2. The high-speed variable clock CKV is passed through a string of non-inverting delay elements, such as buffers. The delayed clock vector D(1:L) is sampled by an array of flip-flops on the rising edge of the reference clock FREF.

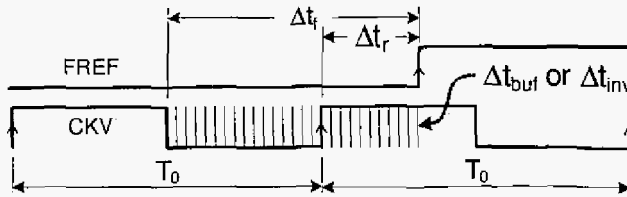


Fig. 3. Timing of the TDC core signals.

The pseudo-thermometer-coded output Q(1:L) contains information on the timing separation between the rising edge of FREF and the rising and falling edges of CKV: Δt_r and Δt_f , respectively, as illustrated in Fig. 3. The output is converted to binary using a simple digital priority decoder [8]. The decoder searches for the first one-to-zero transition that measures the Δt_r CKV-to-FREF delay in units of a buffer delay Δt_{buf} . While this TDC architecture is elegant and simple to implement, it suffers from poor timing resolution due to the selection of the non-inverting delay elements.

It should be recognized that the fastest *logic-level regenerative* timing method in a standard CMOS process is the propagation time of an inverter. Since the non-inverting buffer is composed of two inverters, the inverter-based TDC has a potential to offer improvement in the timing resolution by a factor of two. Interestingly, TDC designs based on the inverter-delay-line have not been reported in the literature so far. The simple elegance of Fig. 2 is plagued there with uneven transition times of the rising and falling edges of the delayed vector signal D(1:L) and unbalanced metastability resolution of the flip-flops. This destroys the even-odd characteristic of the TDC resolution such that the inverter-based delay line TDC offers little advantage over the buffer-based approach.

In Sec. III, we propose a pseudo-differential TDC architecture that realizes the resolution benefits of the inverter-based delay line. In fact, due to the minimized loading on the inverter delay chain, the TDC resolution is very close to the native CMOS inverter propagation delay, which is only 15–20 ps in this CMOS process and further expected to improve in every process technology node advancement.

III. TDC CORE ARCHITECTURE

The TDC core is shown in Fig. 4. Its topology is pseudo-differential to avoid any mismatch between rising and falling edge transitions due to differing strengths of the NMOS and PMOS transistors. To compensate for any phase difference

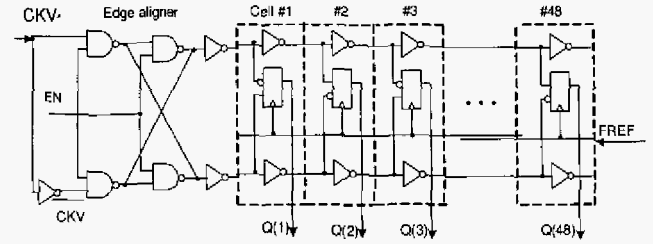


Fig. 4. Implemented TDC core.

in the high-speed complementary clocks (CKV) due to routing mismatches, they are first edge aligned and then passed through a complementary string of $L = 48$ inverters. The edge aligner can tolerate up to 50 ps of skew, so the negating feed can simply be replaced with an inverter. The delayed-clock replica vector is sampled by FREF using an array of 48 sense-amplifier-based flip-flops that are adapted from [10].

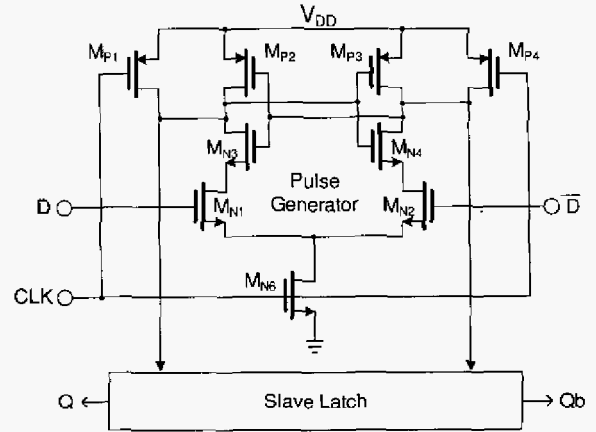


Fig. 5. High-resolution symmetric flip-flop.

The flip-flops (Fig. 5) are symmetrical along the vertical axis and provide identical resolution of the rising and falling edge metastability of their input data. The data input capacitive loading is only one NMOS gate and their interconnect capacitance parasitics are minimized. The metastability window is very small: within a resolution time of 1 ns, the metastability resolution window is less than 1 ps, which is much smaller than the inverter delay. This ensures no “bubbles” in the TDC code and makes the inverter delay the only limitation of the TDC resolution. It should be noted that a typical flip-flop for high-speed digital designs features non-symmetrical metastability characteristic and the composite resolution window exceeds several inverter delays, which makes it unsuitable for this application.

IV. TDC SYSTEM

Fig. 6 shows the TDC core within the larger TDC system. The 48-bit TDC output forms a pseudo-thermometer code,

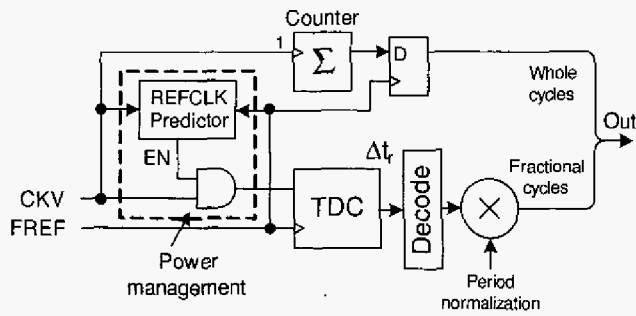


Fig. 6. TDC system with large dynamic range and power saving through clock gating.

which is then converted to binary and normalized to the CKV period, T_0 . The number of inverters is set to cover one T_0 . To arbitrarily increase the dynamic range, the CKV edge counter with a sufficient wordlength is added. The fixed-point TDC output timestamp consists of the sampled CKV edge count (integer part) and the T_0 -normalized delay from CKV to FREF (fractional part). The time difference between the two FREF events is the difference between the two consecutive outputs. In PLL applications, the absolute timestamps (phase) are more useful [7][8] than the time difference (instantaneous frequency). Also, the reference edge locations are quite predictable, so the power is significantly saved by gating off the TDC activity during 90% of the time between the reference edges.

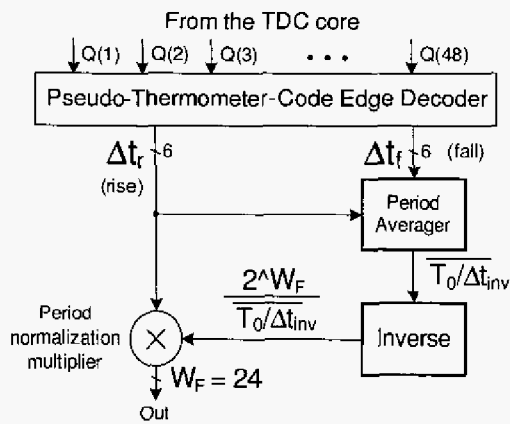


Fig. 7. TDC on-line calibration.

The TDC is also self calibrating during normal operation for the PVT inverter delay variations, as shown in Fig. 7. The absolute difference between the measured rise-edge and fall-edge delays of CKV to FREF (Δt_r and Δt_f , respectively, in Fig. 3) is the half-period of CKV in terms of number of inverters, i.e., half of $T_0/\Delta t_{inv}$. An accurate estimate of $T_0/\Delta t_{inv}$ is obtained through averaging, with an error below 1%. Its inverse is used for the fixed-point period

TABLE 1
COMPARISON TO PREVIOUS WORK.

Work	Technique	CMOS [μm]	Power [mW]	Area [mm^2]	Resol. [ps]
This	pseudo-diff DL	0.09	6.9/1.8	0.01	17–21
[1]	pulse shrinking	0.8	10	6	100
[2]	Vernier DL	0.7	?	10	30
[3]	2-level DL	0.35	50	0.6	24–30
[4]	multi-level DL	0.6	50	<13.6	30

normalization multiplier with the $W_F = 24$ fractional bits. This value divided by the CKV frequency, $1/T_0$, is the inverter delay Δt_{inv} in units of seconds and is read out to give an indication of the CMOS process strength.

V. TDC RESOLUTION EFFECT ON PHASE NOISE

The TDC quantization of timing estimation Δt_{inv} affects the in-band RF output phase noise of the ADPLL of Fig. 1 according to [8]:

$$\mathcal{L} = \frac{(2\pi)^2}{12} \left(\frac{\Delta t_{inv}}{T_0} \right)^2 \cdot \frac{1}{f_R} \quad (1)$$

where, T_0 is the DCO clock period and f_R is the reference or sampling frequency. Substituting $\Delta t_{inv} = 20$ ps, $f_R = 26$ MHz, $f_0 = 1.8$ GHz, $T_0 = 556$ ps, we obtain $\mathcal{L} = -97.8$ dBc/Hz. Eq. (1) was validated experimentally within 1 dB of error for Δt_{inv} spanning 16–34 ps through varying the TDC supply voltage. The in-band phase noise performance provides ample margin for the GSM operation: measured 0.5° of the modulated rms phase noise [9] versus the spec of 5° . The next generations of deep-submicron CMOS processes can only bring reductions in Δt_{inv} , so the phase noise performance will further improve.

VI. EXPERIMENTAL RESULTS

The presented TDC is part of an ADPLL-based single-chip fully-compliant GSM/EDGE transceiver implemented in a digital 90 nm CMOS. The TDC occupies 0.01 mm^2 . The raw current consumption is 5.3 mA at a 1.3 V supply. The clock gating by the power management reduces it to 1.4 mA. The chip implementation and die photo are in [9].

The TDC linearity measurements were carried out using a $f_R = 26$ MHz frequency reference. The $f_0 = 1.6$ –2.0 GHz high-speed clock CKV is generated by the ADPLL by setting the fixed-point frequency ratio $FCW = N = f_0/f_R$ with $W_F = 24$ bits of fractional resolution.

Measured TDC transfer function is shown in Fig. 8. The frequency division ratio N is set slightly off from an integer value so that the FREF and CKV clocks experience slow edge rotation. The TDC output code is averaged out over such 15 cycles to eliminate chatter. The superimposed dotted line is the inverter delay or slope accurately calculated by the calibration hardware. The TDC nonlinearity (Fig. 9) is almost entirely dominated by the even-odd timing mismatch. The measured DNL and INL are both below 0.7 LSB.

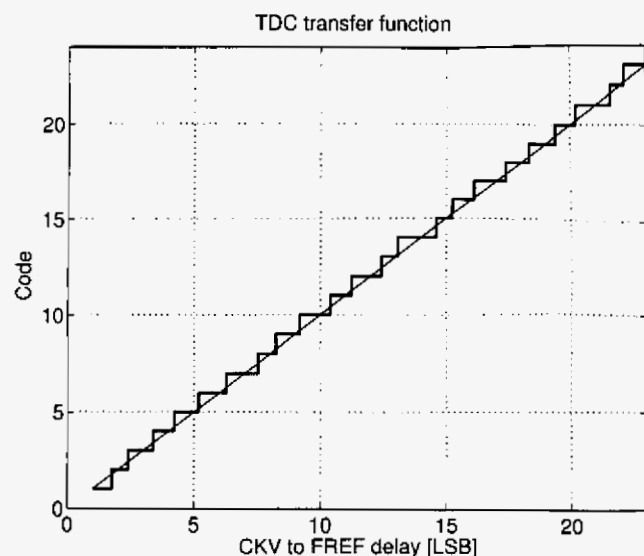


Fig. 8. Measured TDC transfer function.

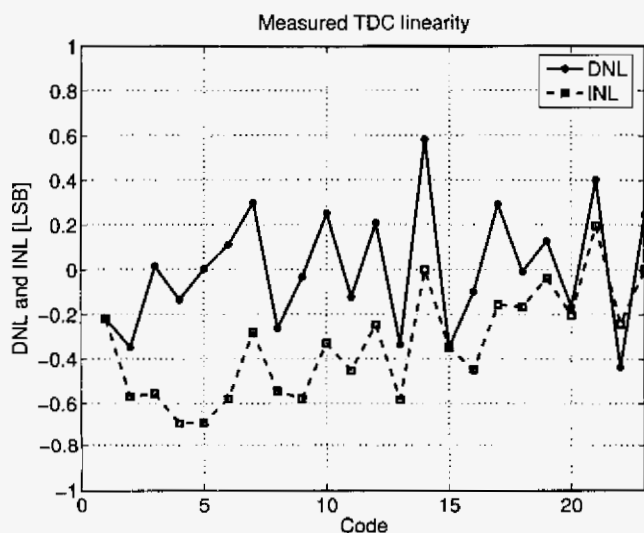


Fig. 9. Measured TDC linearity.

Table I compares the proposed TDC system of Fig. 6 with the prior published work in CMOS technology. It features the fastest resolution despite the simplicity of the fully-digital architecture. The power consumption is an order of magnitude better, whereas the area is two orders of magnitude better, than the prior results. The linearity is also the best, since the prior reported TDC's have INL greater than 1 LSB.

VII. CONCLUSION

We have presented system design, circuit implementation and measurement results of a *time-to-digital converter* (TDC) in a deep-submicron CMOS process. The TDC is specifically employed here as a replacement of a conventional phase/frequency detector and charge pump in a fully-

compliant single-chip GSM transceiver to measure time difference between the PLL oscillator and reference clocks, but it could be used for broad range of applications, such as instrumentation and space science. The TDC core comprises of inverters and flip-flops and does not require any current biasing, analog interpolation or precise scaling of sub-gate delays. The architecture is based on a pseudo-differential topology that makes it less sensitive to differing strengths of NMOS and PMOS transistors. This brings the TDC resolution down to the intrinsic delay of the inverter, instead of the buffer, as in conventional designs. We have also presented an all-digital calibration method of an inverter delay that could be run during the normal operation to compensate for the temperature and voltage drifts. Since the calibration circuit reports a delay of an intrinsic, uncorrected inverter in digital form, it is used as a CMOS process strength indicator for process statistics and to calibrate bias current of other analog circuits on the same die. A novel power management through dynamic clock gating is used in PLL and other applications where the time-of-arrival can be predicted. The measured TDC resolution of 17–21 ps and linearity of INL = 0.7 LSB are the best reported so far in CMOS technology. In addition, the power consumption and area are also the lowest. We expect that in this low-voltage digital CMOS environment, picosecond-level accurate time interval measurements will become as important as high-resolution A/D conversion in conventional architectures.

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